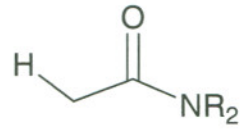
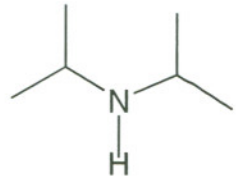
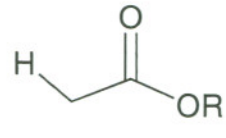
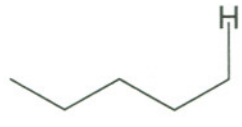
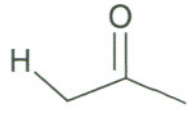
Organic Division Cumulative Exam

October 19, 2007

1. (a) On the left side of the page, draw the **complete** set of group MO's for planar CH_3 using the H and C s and p atomic orbitals. Draw them so that the lowest energy MO is at the bottom of the diagram. Indicate phasing by shading in the appropriate orbitals. Keep in mind that there are 2 sets of 2 degenerate MO's. (b) In the middle of the page, draw the **complete** set of group MO's for pyramidal CH_3. (c) On the right side of the page, draw any necessary hybrid orbitals for pyramidal CH_3. (d) Write the names of the 4 lowest energy MO's next to the drawings.

2. (a) Using the relevant MO's from problem #1, draw the MO's of **staggered** ethane up to and including the LUMO. (b) Indicate their relative energies and designate each MO with the appropriate name and sign (+/-). In order to have the correct relative energies, you will only need to know that the LUMO is the set of 2 degenerate ethane MO's. (c) Designate the HOMO and LUMO. (d) Fill in the number of electrons that corresponds to neutral ethane.

3. Given two equilibrating species A and B, what energy difference in kcal/mol does there have to be between A and B to afford a 10:1 ratio of A:B at 298 K?

4. What is the approximate vicinal coupling constant between two protons which possess an H-C-C-H dihedral angle of 180°? Of 0 °?

5. Rank the acidity from most to least acidic and give the approximate pK_a's of the following compounds.